Day 123 | Unit 09 | Teacher plan

Display, Describe, and Summarize Data

Display and Interpret Dot Plots

Standards: 6.SP.B.4

Learning target: I can construct and interpret a dot plot on a number line.

enVision Topic 8: Display, Describe, and Summarize Data | Spectrum companion Chapter 7

0–5 minutes / Launch

Read the target. Ask for an estimate or a sketch before calculating. Connect the new idea to the previous lesson.

5–17 / Model

Use Worked Example 1, then Example 2. Ask students to name the next step and explain a choice. Draw an equally spaced number line and stack one dot per observation above its value. Repeated values stack vertically. Add an axis label with units and check that the dot count equals the data count.

17–42 / Practice

Assign questions 1–10 in order. Use 1–3 for foundations, 4–7 for application, and 8–10 for state-format response and reasoning. The example button provides a separate question with a worked method.

42–50 / Discuss and check

Compare representations. Review a selected answer and a plausible error. Use question 10 as an exit discussion; ask students to check units, scale, or substitution.

Support and extension

Offer number lines, fraction strips, grid paper, or a table template as appropriate. Read multi-part directions one part at a time. For extension, ask for another representation or a counterexample.

Companion and state alignment

Use the matching skill in the edition you own. No unverified publisher page numbers are assigned. AASA references below identify related public sample items by standard and format; all course questions are original.

AASA sample item references: 13

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01 / Learn

Draw an equally spaced number line and stack one dot per observation above its value. Repeated values stack vertically. Add an axis label with units and check that the dot count equals the data count.

Worked example 1

Reading times in minutes are 7, 8, 8, 9, 9, 10. Draw a labeled dot plot. Check the number of dots.

1. Draw an equally spaced number line and stack one dot per observation above its value.

2. Repeated values stack vertically.

3. Each observation contributes one dot

4. stack repeated values at their number-line position.

5. State the result in the form requested: 7: 1 dot; 8: 2 dots; 9: 2 dots; 10: 1 dot; n = 6.

Answer: 7: 1 dot; 8: 2 dots; 9: 2 dots; 10: 1 dot; n = 6

Worked example 2

A frequency table records books read: 0 books: 1 students; 1: 5; 2: 4; 3: 1. Part A: draw a dot plot. Part B: enter the observation count and explain how to verify it.

1. Draw an equally spaced number line and stack one dot per observation above its value.

2. Place one dot per student, not one per distinct value.

3. Total dots = 1 + 5 + 4 + 1 = 11.

4. Label the number line books read.

Answer: 11 students; stacks at 0, 1, 2, 3 have 1, 5, 4, 1 dots.

Day 123 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Reading times in minutes are 8, 9, 9, 10, 10, 11. Draw a labeled dot plot. Check the number of dots.

02 / Constructed response

Reading times in minutes are 2, 3, 3, 4, 4, 5. Draw a labeled dot plot. Check the number of dots.

03 / Constructed response

Reading times in minutes are 4, 5, 5, 6, 6, 7. Draw a labeled dot plot. Check the number of dots.

04 / Constructed response

Reading times in minutes are 4, 6, 6, 8, 8, 10. Draw a labeled dot plot. Check the number of dots.

05 / Constructed response

Reading times in minutes are 3, 8, 8, 13, 13, 18. Draw a labeled dot plot. Check the number of dots.

06 / Constructed response

Reading times in minutes are 5, 11, 11, 17, 17, 23. Draw a labeled dot plot. Check the number of dots.

07 / Constructed response

Reading times in minutes are 5, 6, 6, 7, 8, 17. Draw a labeled dot plot. Describe a cluster or gap and report the count.

08 / Data display and numeric entry

A frequency table records books read: 0 books: 3 students; 1: 5; 2: 3; 3: 1. Part A: draw a dot plot. Part B: enter the observation count and explain how to verify it.



Choose and label an equal-interval scale.

09 / Data display and numeric entry

A frequency table records books read: 0 books: 4 students; 1: 4; 2: 5; 3: 5. Part A: draw a dot plot. Part B: enter the observation count and explain how to verify it.



Choose and label an equal-interval scale.

10 / Constructed response

Reading times in minutes are 6, 7, 7, 8, 9, 18. Draw a labeled dot plot. Describe a cluster or gap and report the count.

Day 123 | Unit 09 | Answer key

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1. 8: 1 dot; 9: 2 dots; 10: 2 dots; 11: 1 dot; n = 6

Each observation contributes one dot; stack repeated values at their number-line position.

6.SP.B.4 | Constructed response

2. 2: 1 dot; 3: 2 dots; 4: 2 dots; 5: 1 dot; n = 6

Each observation contributes one dot; stack repeated values at their number-line position.

6.SP.B.4 | Constructed response

3. 4: 1 dot; 5: 2 dots; 6: 2 dots; 7: 1 dot; n = 6

Each observation contributes one dot; stack repeated values at their number-line position.

6.SP.B.4 | Constructed response

4. 4: 1 dot; 6: 2 dots; 8: 2 dots; 10: 1 dot; n = 6

Each observation contributes one dot; stack repeated values at their number-line position.

6.SP.B.4 | Constructed response

5. 3: 1 dot; 8: 2 dots; 13: 2 dots; 18: 1 dot; n = 6

Each observation contributes one dot; stack repeated values at their number-line position.

6.SP.B.4 | Constructed response

6. 5: 1 dot; 11: 2 dots; 17: 2 dots; 23: 1 dot; n = 6

Each observation contributes one dot; stack repeated values at their number-line position.

6.SP.B.4 | Constructed response

7. 5: 1 dot; 6: 2 dots; 7: 1 dot; 8: 1 dot; 17: 1 dot; n = 6; low cluster and high isolated value

Each observation contributes one dot; stack repeated values at their number-line position.

6.SP.B.4 | Constructed response

8. 12 students; stacks at 0, 1, 2, 3 have 3, 5, 3, 1 dots.

Place one dot per student, not one per distinct value. Total dots = 3 + 5 + 3 + 1 = 12. Label the number line books read.

6.SP.B.4 | Data display and numeric entry

9. 18 students; stacks at 0, 1, 2, 3 have 4, 4, 5, 5 dots.

Place one dot per student, not one per distinct value. Total dots = 4 + 4 + 5 + 5 = 18. Label the number line books read.

6.SP.B.4 | Data display and numeric entry

10. 6: 1 dot; 7: 2 dots; 8: 1 dot; 9: 1 dot; 18: 1 dot; n = 6; low cluster and high isolated value

Each observation contributes one dot; stack repeated values at their number-line position.

6.SP.B.4 | Constructed response